

SHOP CUSTOMER DATA

Kaggle Data Analysis Project

Python • Pandas • Matplotlib • Seaborn

1. Project Objective

Analyze shop customer information using Python.

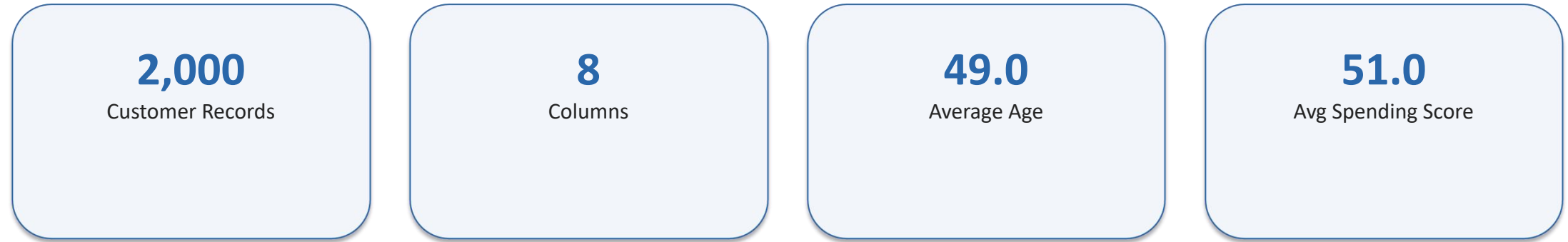
Understand customer demographics, income and spending behavior.

Inspect missing values and basic data quality.

Create clear visualizations to find useful patterns.

Prepare the dataset for future customer segmentation or machine-learning work.

2. Dataset Overview



The dataset contains demographic and customer-behavior information.

Main fields include Gender, Age, Annual Income, Spending Score, Profession, Work Experience and Family Size.

3. Dataset Columns

CustomerID — Unique customer identifier

Gender — Customer gender

Age — Customer age

Annual Income (\$) — Annual income

Spending Score (1-100) — Customer spending score

Profession — Customer profession

Work Experience — Years of work experience

Family Size — Family size

4. Tools & Libraries

Python — programming language for the analysis.

Pandas — loading, cleaning, filtering and analyzing tabular data.

Matplotlib — creating charts and histograms.

Seaborn — creating statistical and categorical visualizations.

Kaggle Notebook — environment used to run and present the analysis.

5. Data Inspection & Quality

The notebook checks the first rows of the dataset.

Dataset shape and column names are inspected.

Data types and missing values are checked.

Descriptive statistics are generated with `df.describe()`.

Profession contains 35 missing values.

These missing values should be handled before future machine-learning modeling.

6. Basic Customer Statistics

48.96 years

Average Age

\$110,732

Average Income

50.96

Average Spending

3.77

Avg Family Size

These averages provide a quick overview of the customer population.

They can be used as a baseline for deeper customer analysis.

7. Analysis & Visualizations

Gender distribution — compares the number of customers by gender.

Profession distribution — shows how customers are distributed across professions.

Age distribution — shows the spread of customer ages.

Income vs Spending Score — explores the relationship between income and spending.

Correlation heatmap — shows relationships between numerical variables.

Top 10 customers — identifies customers with the highest spending scores.

8. Income vs Spending Score

A scatter plot is used to compare Annual Income with Spending Score.

Each point represents a customer.

Gender is used to distinguish customers visually.

This visualization can help identify customer patterns and possible segments.

A future project can use these features for customer segmentation.

9. Conclusion

The project successfully performs exploratory analysis of shop customer data.

Python libraries make it easier to inspect, summarize and visualize customer information.

The dataset contains useful demographic, income and spending features.

Missing Profession values should be handled carefully before advanced modeling.

The project can be extended into customer segmentation and machine-learning applications.

10. Future Improvements

Clean and handle missing Profession values.

Create customer segments using income and spending score.

Compare spending behavior across professions and age ranges.

Study the effect of work experience and family size on spending.

Build a machine-learning model for customer segmentation or prediction.

THANK YOU

Shop Customer Data — Kaggle Project